

Imp imports

Data handling

```
import pandas as pd      # DataFrames, CSV/Excel I/O, groupby, merge
import numpy as np       # Arrays, numerical ops, linear algebra, random
```

Plotting / EDA

```
import matplotlib.pyplot as plt # Basic plotting (use plt.figure(), plt.plot(), plt.hist())
import seaborn as sns          # High-level stats plots (pairplot, heatmap, boxplot)
```

Train / test split

```
from sklearn.model_selection import train_test_split # split dataset into train/test
```

Preprocessing

```
from sklearn.preprocessing import StandardScaler, MinMaxScaler # scale features
from sklearn.preprocessing import LabelEncoder, OneHotEncoder # categorical
encoding
from sklearn.impute import SimpleImputer          # missing value handling
from sklearn.pipeline import Pipeline              # chain steps
from sklearn.compose import ColumnTransformer      # separate numeric / cat
pipelines
```

Models (examples)

```
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.decomposition import PCA
```

```
# Model selection / tuning

from sklearn.model_selection import GridSearchCV, cross_val_score

from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, roc_auc_score, mean_squared_error,
                             r2_score, confusion_matrix, classification_report)
```